

II-Probabilistic Approaches Independence and Graphical Models

Foundations of Neuro-Symbolic AI

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How to beat the curse of dimensions?

Probability distribution of factored systems with states $\times_{k \in [d]} [m_k]$ has

$$\left(\prod_{k \in [d]} m_k \right) - 1$$

degrees of freedom (coordinates to specify and store).

Mitigation: [Tensor Network Decompositions](#)

Independencies of Random Variables

Decompositions of Probability Tensors correspond with independencies of (hidden) random variables.

Example: Being at a dentist

Add a variable **Cloud**, denoting the weather outside the dentists lab.

- ▶ This adds an additional axis to $\mathbb{P}$, thus the number of coordinates increases by a factor of 2.
- ▶ But: Intuitively, knowing **Cloud** should not affect the probability of having a cavity, so why shall we care?

Independence of Cloud to the other Variables

After showing that cavity, catch and toothache are independent of cloud, we do not have to consider cloud any more.

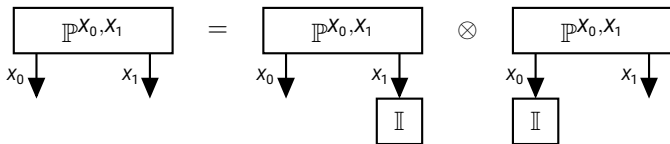
Formal definition of Independencies

Definition (Independence)

Given a joint distribution of variables X_0 and X_1 , we say that X_0 is independent from X_1 if for any values x_{X_0}, x_{X_1} we have

$$\mathbb{P}^{X_0=x_{X_0}, X_1=x_{X_1}} = \mathbb{P} [X_0 = x_{X_0}] \cdot \mathbb{P} [X_1 = x_{X_1}].$$

In the tensor network decomposition we depict this by



Contraction criterion for independence

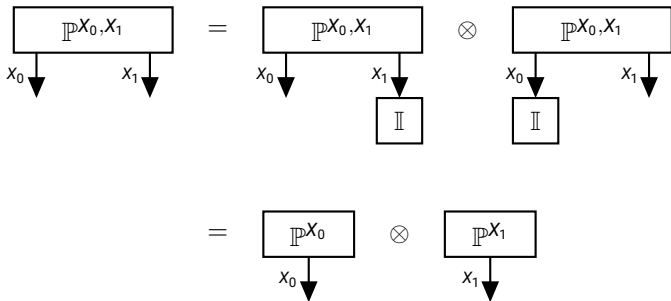
Theorem

Given a probability distribution $\mathbb{P}$, X_0 is independent from X_1 , if and only if

$$\mathbb{P} = \langle \mathbb{P} \rangle_{[X_0]} \otimes \langle \mathbb{P} \rangle_{[X_1]} .$$

Decomposition into Marginal Probability Tensors

Independence allows the decomposition into



Exponential to linear storage demand

Instead of storing $m_{X_0} \cdot m_{X_1}$ coordinates, we can store $\mathbb{P}$ with $m_{X_0} + m_{X_1}$ demand.

Formal definition of Conditional Independencies

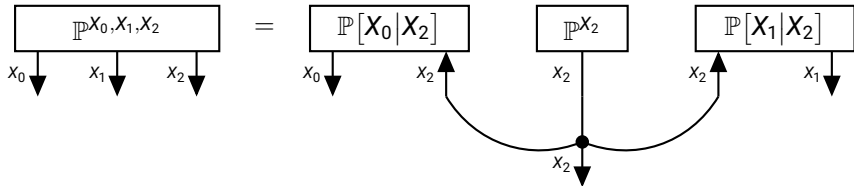
Definition (Conditional Independence)

Given a joint distribution of variables X_0 , X_1 and X_2 , we say X_0 is independent from X_1 conditioned on X_2 if

$$\mathbb{P}[X_0, X_1 | X_2] = \mathbb{P}[X_0 | X_2] \cdot \mathbb{P}[X_1 | X_2] .$$

Decomposition given conditional independence

We depict conditional independence by tensor network decompositions:



Chain Rule: Decomposing Probabilities

Theorem (Chain Rule)

For any joint probability distribution $\mathbb{P}$ of the variables $\mathbb{P}^{X_0, \dots, X_{d-1}}$ we have

$$\mathbb{P} = \langle \{ \mathbb{P}[X_k | X_0, \dots, X_{k-1}] : k \in [d] \} \rangle_{[\{f^0, \dots, f^{d-1}\}]}$$

where for $k = 1$ we denote by $\mathbb{P}[X_0 | X_0, \dots, X_{-1}]$ the marginal distribution $\mathbb{P}^{X_0}$.

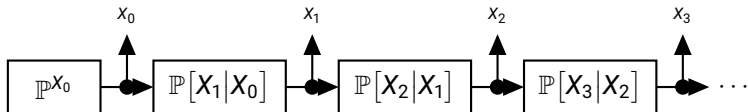
Markov Chains

Theorem (Markov Chain)

Let there be a set of variables X_t where $t \in [T]$. When X_t is independent of $X_{0:t-2}$ conditioned on X_{t-1} (the Markov Property), then

$$\mathbb{P} = \langle \mathbb{P}[X_t | X_{t-1}] : t \in [T] \rangle_{[X_0, \dots, X_{T-1}]} .$$

We depict this by



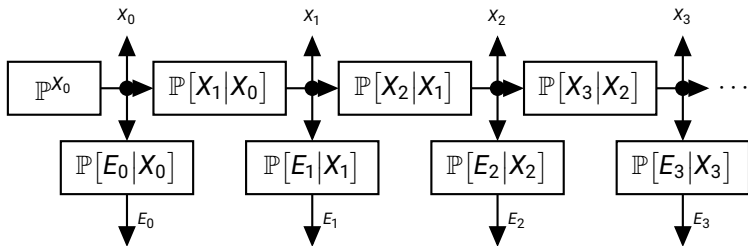
Hidden Markov Models

Hidden Markov Models extend Markov Chains by limited observation E_t of the variables X_t .

The independence assumptions are

- ▶ X_{t+1} is independent of $X_{0:t-1}$ and $E_{0:t}$ conditioned on X_t
- ▶ E_t is independent of all other variables conditioned on X_t

The independence assumptions are exploited in the decomposition



Directed Graphical Models: Bayesian Networks

Definition (Bayesian Networks)

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed acyclic graph and for each node $v \in \mathcal{V}$ a random variable X_v . Further let there be for each node $v \in \mathcal{V}$ with parents $\text{Pa}(v)$ a conditional probability distribution

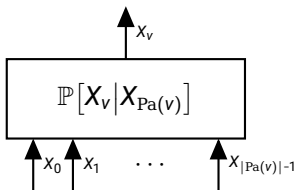
$$\mathbb{P}[X_v | X_{\text{Pa}(v)}] .$$

Then the **Bayesian Network** with respect to $\mathcal{G}$ and the conditional probability terms is the distribution

$$\mathbb{P} = \langle \mathbb{P}[X_v | X_{\text{Pa}(v)}] : v \in \mathcal{V} \rangle_{[X_v : v \in \mathcal{V}]} .$$

Directed Graphical Models: Bayesian Networks

For each variable we build the conditional probability tensor



The **Bayesian Network** is then the contraction

$$\mathbb{P} = \langle \mathbb{P}[X_v | X_{\text{Pa}(v)}] : v \in \mathcal{V} \rangle_{[X_v : v \in \mathcal{V}]} .$$

Undirected Graphical Models: Markov Networks

Definition (Markov Networks)

Let $\tau^{\mathcal{G}}$ be a Tensor Network on a hypergraph $\mathcal{G}$. The associated Markov Network is the probability distribution of $\{X_v : v \in \mathcal{V}\}$ defined by

$$\mathbb{P}^{\mathcal{G}} = \frac{\langle \{\tau^e : e \in \mathcal{E}\} \rangle_{[\mathcal{V}]}}{\langle \{\tau^e : e \in \mathcal{E}\} \rangle_{[\emptyset]}} = \langle \{\tau^e : e \in \mathcal{E}\} \rangle_{[\mathcal{V}|\emptyset]} .$$

We call the denominator

$$\mathcal{Z}(\mathcal{G}) = \langle \{\tau^e : e \in \mathcal{E}\} \rangle_{[\emptyset]}$$

the **partition function** of the Markov Network.